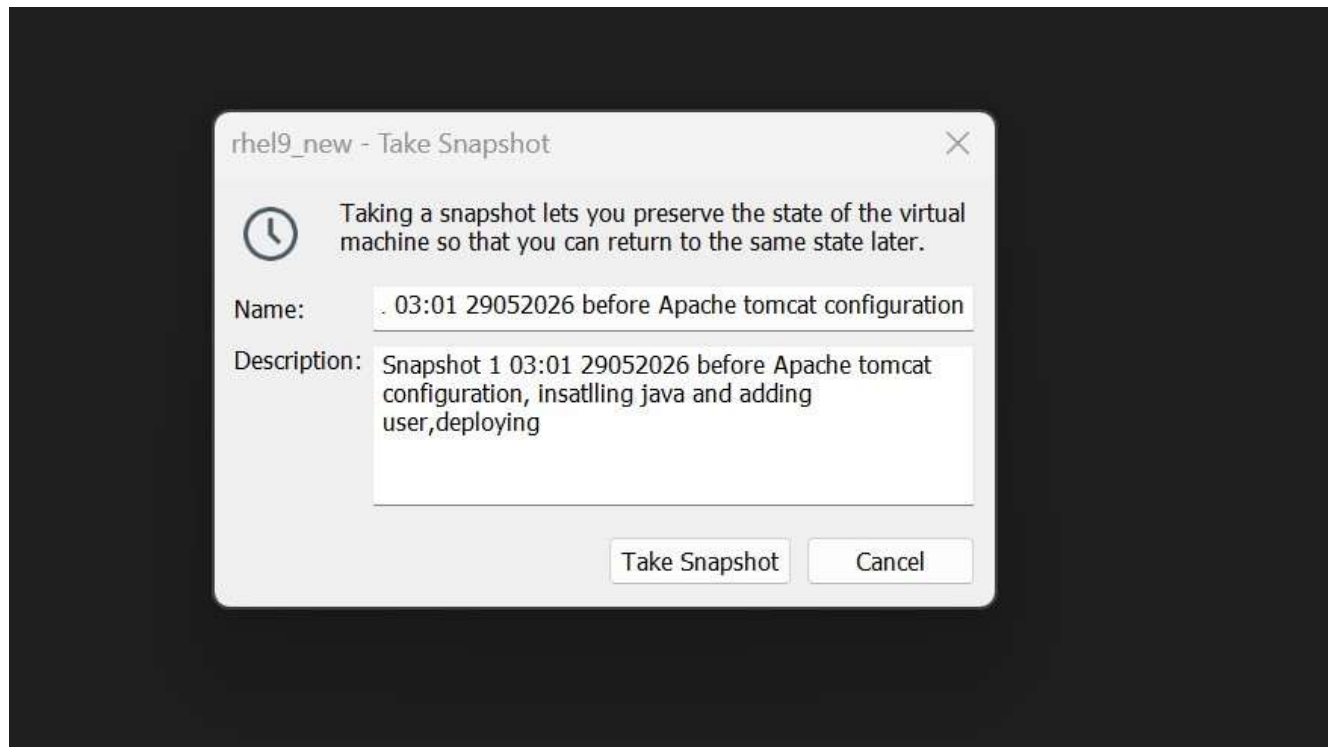


[Type here]

Take snapshot



2) set IP to NAT(DHCP)

```
root@localhost:~  
[root@localhost ~]# ifconfig  
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.80.134 netmask 255.255.255.0 broadcast 192.168.80.255  
    inet6 fe80::20c:29ff:fed6:15d0 prefixlen 64 scopeid 0x20<link>  
    ether 00:0c:29:d6:15:d0 txqueuelen 1000 (Ethernet)  
    RX packets 246 bytes 290313 (283.5 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 168 bytes 13870 (13.5 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 17 bytes 2062 (2.0 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 17 bytes 2062 (2.0 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

[Type here]

3) since dns server is not configured so we have put entry in /etc/hosts

```
[root@tomcatsrv1 ~]#  
[root@tomcatsrv1 ~]# vi /etc/hosts  
[root@tomcatsrv1 ~]#  
[root@tomcatsrv1 ~]# cat /etc/hosts  
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4  
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6  
  
192.168.80.134 tomcatsrv1.radical.com  
[root@tomcatsrv1 ~]#
```

4)check name resolution & Internet connectivity

```
[root@tomcatsrv1 ~]# ping -c2 tomcatsrv1.radical.com  
PING tomcatsrv1.radical.com (192.168.80.134) 56(84) bytes of data.  
64 bytes from tomcatsrv1.radical.com (192.168.80.134): icmp_seq=1 ttl=64 time=0.148 ms  
64 bytes from tomcatsrv1.radical.com (192.168.80.134): icmp_seq=2 ttl=64 time=0.137 ms  
  
--- tomcatsrv1.radical.com ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1056ms  
rtt min/avg/max/mdev = 0.137/0.142/0.148/0.005 ms  
[root@tomcatsrv1 ~]#  
[root@tomcatsrv1 ~]# ping -c2 www.google.com  
PING www.google.com (142.251.150.119) 56(84) bytes of data.  
64 bytes from 142.251.150.119 (142.251.150.119): icmp_seq=1 ttl=128 time=17.5 ms  
64 bytes from 142.251.150.119 (142.251.150.119): icmp_seq=2 ttl=128 time=13.5 ms  
  
--- www.google.com ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1002ms  
rtt min/avg/max/mdev = 13.457/15.472/17.488/2.015 ms  
[root@tomcatsrv1 ~]#  
[root@tomcatsrv1 ~]# ping -c2 www.redhat.com  
PING e3396.dscx.akamaiedge.net (23.38.58.41) 56(84) bytes of data.  
64 bytes from a23-38-58-41.deploy.static.akamaitechnologies.com (23.38.58.41): icmp_seq=1 ttl=128 time=23.3 ms  
64 bytes from a23-38-58-41.deploy.static.akamaitechnologies.com (23.38.58.41): icmp_seq=2 ttl=128 time=12.7 ms  
  
--- e3396.dscx.akamaiedge.net ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1003ms  
rtt min/avg/max/mdev = 12.724/17.991/23.258/5.267 ms  
[root@tomcatsrv1 ~]#
```

[Type here]

5)Configure Required Repository

```
root@localhost:~# yum repolist all
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

repo id                                repo name                                status
AppStream                              Redhat-AppStream                        enabled
BaseOS                                  Redhat-BaseOS                          enabled
[root@tomcatsrv1 ~]#
```

6)Install java package

```
root@localhost:~# yum search java-11-openjdk
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

Last metadata expiration check: 0:09:24 ago on Fri 29 May 2026 03:33:57 PM IST.
===== Name Exactly Matched: java-11-openjdk =====
java-11-openjdk.x86_64 : OpenJDK 11 Runtime Environment
===== Name Matched: java-11-openjdk =====
java-11-openjdk-demo.x86_64 : OpenJDK 11 Demos
java-11-openjdk-devel.x86_64 : OpenJDK 11 Development Environment
java-11-openjdk-headless.x86_64 : OpenJDK 11 Headless Runtime Environment
java-11-openjdk-javadoc.x86_64 : OpenJDK 11 API documentation
java-11-openjdk-javadoc-zip.x86_64 : OpenJDK 11 API documentation compressed in a single archive
java-11-openjdk-jmods.x86_64 : JMods for OpenJDK 11
java-11-openjdk-src.x86_64 : OpenJDK 11 Source Bundle
java-11-openjdk-static-libs.x86_64 : OpenJDK 11 libraries for static linking
[root@tomcatsrv1 ~]#
[root@tomcatsrv1 ~]#
[root@tomcatsrv1 ~]# yum install -y java-11-openjdk
```

[Type here]

6) Creates a **system user named tomcat** with a home directory `/opt/tomcat`, no login shell (`/bin/false`), and a matching group, mainly for securely running the Tomcat service.

```
root@localhost:~  
[root@tomcatsrv1 ~]# useradd -r -m -U -d /opt/tomcat -s /bin/false tomcat  
C  
[root@tomcatsrv1 ~]# cat /etc/passwd | grep tomcat  
tomcat:x:978:977::/opt/tomcat:/bin/false  
[root@tomcatsrv1 ~]#
```

7) remove if tomcat dir is already there in /opt/

```
root@localhost:~  
[root@tomcatsrv1 ~]# cd /opt/  
[root@tomcatsrv1 opt]#  
[root@tomcatsrv1 opt]# ls  
tomcat  
[root@tomcatsrv1 opt]# rm -rvf tomcat/  
removed directory 'tomcat/.mozilla/extensions'  
removed directory 'tomcat/.mozilla/plugins'  
removed directory 'tomcat/.mozilla'  
removed 'tomcat/.bash_logout'  
removed 'tomcat/.bash_profile'  
removed 'tomcat/.bashrc'  
removed directory 'tomcat/'  
[root@tomcatsrv1 opt]#  
[root@tomcatsrv1 opt]#
```

[Type here]

8)search stable and latest version of tomcat

<https://downloads01-he-fi.apache.org/tomcat/tomcat-10/v10.1.55/bin/apache-tomcat-10.1.55.tar.gz>

← ↻ 🔒 https://downloads01-he-fi.apache.org/tomcat/tomcat-10/v10.1.55/bin/

Parent Directory	-	
 embed/	2026-05-11 21:20	-
 apache-tomcat-10.1.55-deployer.tar.gz	2026-05-05 13:39	2.8M
 apache-tomcat-10.1.55-deployer.tar.gz.asc	2026-05-05 13:39	833
 apache-tomcat-10.1.55-deployer.tar.gz.sha512	2026-05-05 13:39	167
 apache-tomcat-10.1.55-deployer.zip	2026-05-05 13:39	2.9M
 apache-tomcat-10.1.55-deployer.zip.asc	2026-05-05 13:39	833
 apache-tomcat-10.1.55-deployer.zip.sha512	2026-05-05 13:39	164
 apache-tomcat-10.1.55-fulldocs.tar.gz	2026-05-05 13:39	36M
 apache-tomcat-10.1.55-fulldocs.tar.gz.asc	2026-05-05 13:39	833
 apache-tomcat-10.1.55-fulldocs.tar.gz.sha512	2026-05-05 13:39	167
 apache-tomcat-10.1.55-windows-x64.zip	2026-05-05 13:39	16M
 apache-tomcat-10.1.55-windows-x64.zip.asc	2026-05-05 13:39	833
 apache-tomcat-10.1.55-windows-x64.zip.sha512	2026-05-05 13:39	167
 apache-tomcat-10.1.55-windows-x86.zip	2026-05-05 13:39	15M
 apache-tomcat-10.1.55-windows-x86.zip.asc	2026-05-05 13:39	833
 apache-tomcat-10.1.55-windows-x86.zip.sha512	2026-05-05 13:39	167
 apache-tomcat-10.1.55.exe	2026-05-05 13:39	15M
 apache-tomcat-10.1.55.exe.asc	2026-05-05 13:39	833
 apache-tomcat-10.1.55.exe.sha512	2026-05-05 13:39	155
 apache-tomcat-10.1.55.tar.gz	2026-05-05 13:39	14M
 apache-tomcat-10.1.55.tar.gz.asc	2026-05-05 13:39	833
 apache-tomcat-10.1.55.tar.gz.sha512	2026-05-05 13:39	158
 apache-tomcat-10.1.55.zip	2026-05-05 13:39	14M
 apache-tomcat-10.1.55.zip.asc	2026-05-05 13:39	833

https://downloads01-he-fi.apache.org/tomcat/tomcat-10/v10.1.55/bin/apache-tomcat-10.1.55-fulldocs.tar.gz

[Type here]

9) wget in /opt for third-party software

```
root@localhost:/opt
[root@tomcatsrv1 opt]# wget https://downloads01-he-fi.apache.org/tomcat/tomcat-10/v10.1.55/bin/apache-tomcat-10.1.55.tar.gz
--2026-05-29 16:02:50-- https://downloads01-he-fi.apache.org/tomcat/tomcat-10/v10.1.55/bin/apache-tomcat-10.1.55.tar.gz
Resolving downloads01-he-fi.apache.org (downloads01-he-fi.apache.org)... 95.216.224.44, 2a01:4f9:2b:1cc2::2
Connecting to downloads01-he-fi.apache.org (downloads01-he-fi.apache.org)|95.216.224.44|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14343328 (14M) [application/x-gzip]
Saving to: 'apache-tomcat-10.1.55.tar.gz'

apache-tomcat-10.1.55.tar.gz 100%[=====] 13.68M 903KB/s in 19s

2026-05-29 16:03:10 (729 KB/s) - 'apache-tomcat-10.1.55.tar.gz' saved [14343328/14343328]

[root@tomcatsrv1 opt]# ls -lrt
total 14008
-rw-r--r--. 1 root root 14343328 May 5 19:09 apache-tomcat-10.1.55.tar.gz
[root@tomcatsrv1 opt]#
```

[Type here]

10)create dir /opt/tomcat and extract tar file and give required permission-

```
[root@tomcatsrv1 opt]# mkdir /opt/tomcat
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# tar -xzf apache-tomcat-10.1.55.
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# chown -R tomcat:tomcat /opt/tom
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# ll
total 14012
-rw-r--r--. 1 root    root    14343328 May  5 19:09 apac
drwxr-xr-x. 9 tomcat  tomcat    4096 May 29 16:07 tomc
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# ls -l /opt/tomcat/
total 144
drwxr-x---. 2 tomcat tomcat  4096 May 29 16:07 bin
-rw-r-----. 1 tomcat tomcat 26826 May  5 19:03 BUILDING
drwx-----. 2 tomcat tomcat  4096 May  5 19:03 conf
-rw-r-----. 1 tomcat tomcat  8626 May  5 19:03 CONTRIB
drwxr-x---. 2 tomcat tomcat  4096 May 29 16:07 lib
-rw-r-----. 1 tomcat tomcat 60393 May  5 19:03 LICENSE
drwxr-x---. 2 tomcat tomcat    6 May  5 19:03 logs
-rw-r-----. 1 tomcat tomcat  2333 May  5 19:03 NOTICE
-rw-r-----. 1 tomcat tomcat  3231 May  5 19:03 README.
-rw-r-----. 1 tomcat tomcat  6776 May  5 19:03 RELEASE
-rw-r-----. 1 tomcat tomcat 16114 May  5 19:03 RUNNING
drwxr-x---. 2 tomcat tomcat   30 May 29 16:07 temp
drwxr-x---. 7 tomcat tomcat   81 May  5 19:03 webapps
drwxr-x---. 2 tomcat tomcat    6 May  5 19:03 work
[root@tomcatsrv1 opt]#
```

[Type here]

Used to assign ownership of Tomcat files to the `tomcat` user for secure service management
Used to provide execute permission to all Tomcat startup and shutdown shell scripts.

```
root@localhost:/opt
[root@tomcatsrv1 opt]# sh -c 'chmod +x /opt/tomcat/bin/*.sh'
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# ll tomcat/bin/
total 936
-rw-r-----. 1 tomcat tomcat 36412 May  5 19:03 bootstrap.jar
-rw-r-----. 1 tomcat tomcat 15777 May  5 19:03 catalina.bat
-rwxr-x--x. 1 tomcat tomcat 24049 May  5 19:03 catalina.sh
-rw-r-----. 1 tomcat tomcat 1664 May  5 19:03 catalina-tasks.xml
-rw-r-----. 1 tomcat tomcat 2123 May  5 19:03 ciphers.bat
-rwxr-x--x. 1 tomcat tomcat 1997 May  5 19:03 ciphers.sh
-rw-r-----. 1 tomcat tomcat 26901 May  5 19:03 commons-daemon.jar
-rw-r-----. 1 tomcat tomcat 218507 May  5 19:03 commons-daemon-native.tar.gz
-rw-r-----. 1 tomcat tomcat 2040 May  5 19:03 configtest.bat
-rwxr-x--x. 1 tomcat tomcat 1922 May  5 19:03 configtest.sh
-rwxr-x--x. 1 tomcat tomcat 8319 May  5 19:03 daemon.sh
-rw-r-----. 1 tomcat tomcat 2091 May  5 19:03 digest.bat
-rwxr-x--x. 1 tomcat tomcat 1965 May  5 19:03 digest.sh
-rw-r-----. 1 tomcat tomcat 3606 May  5 19:03 makebase.bat
-rwxr-x--x. 1 tomcat tomcat 3382 May  5 19:03 makebase.sh
-rw-r-----. 1 tomcat tomcat 2096 May  5 19:03 migrate.bat
-rwxr-x--x. 1 tomcat tomcat 1970 May  5 19:03 migrate.sh
-rw-r-----. 1 tomcat tomcat 3382 May  5 19:03 setclasspath.bat
-rwxr-x--x. 1 tomcat tomcat 3941 May  5 19:03 setclasspath.sh
-rw-r-----. 1 tomcat tomcat 2020 May  5 19:03 shutdown.bat
-rwxr-x--x. 1 tomcat tomcat 1902 May  5 19:03 shutdown.sh
-rw-r-----. 1 tomcat tomcat 2022 May  5 19:03 startup.bat
-rwxr-x--x. 1 tomcat tomcat 1904 May  5 19:03 startup.sh
-rw-r-----. 1 tomcat tomcat 52448 May  5 19:03 tomcat-juli.jar
```

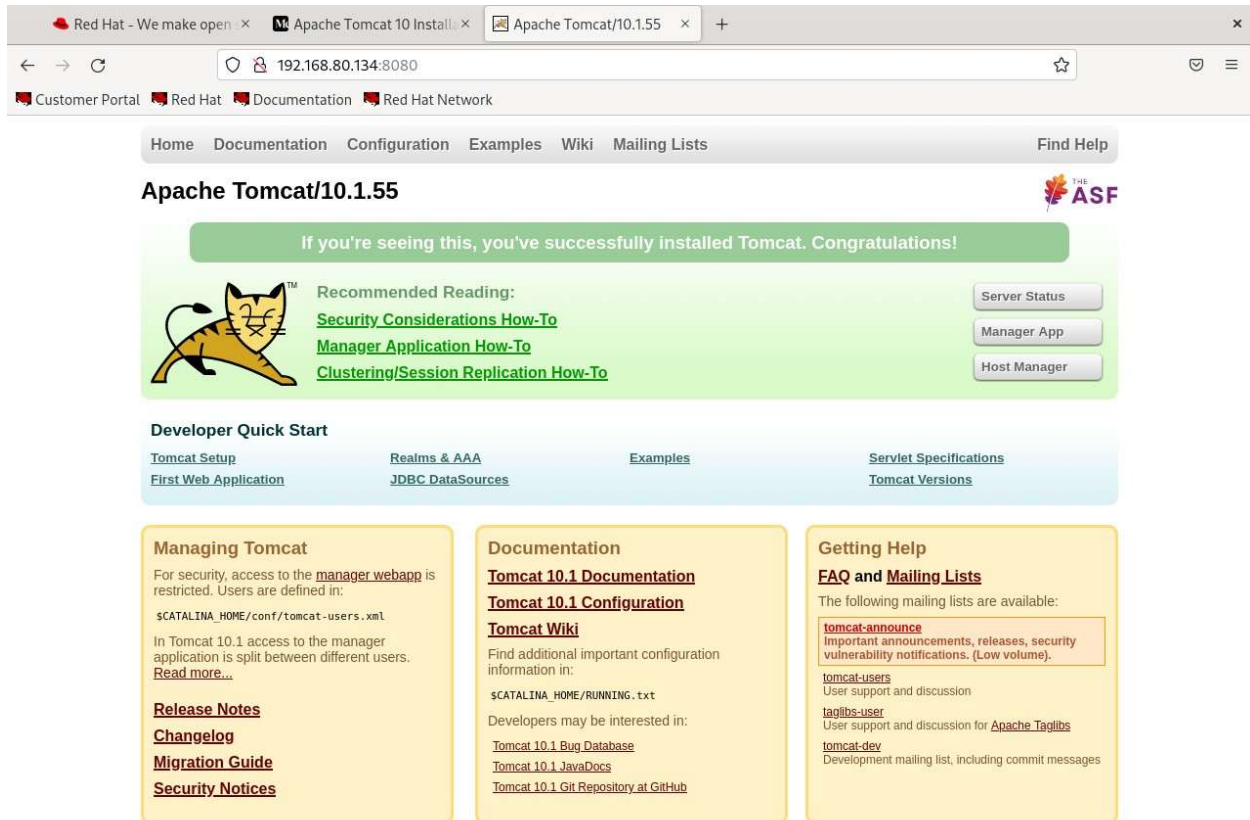
11) startup.sh & shutdown.sh file is use to start and stop the service
when tomcat.service file is not available

```
root@localhost:/opt
[root@tomcatsrv1 opt]# ls tomcat/bin/
bootstrap.jar      commons-daemon.jar      digest.sh           setclasspath.sh     tomcat-native.tar.gz
catalina.bat      commons-daemon-native.tar.gz  makebase.bat       shutdown.bat        tool-wrapper.bat
catalina.sh       configtest.bat          makebase.sh        shutdown.sh          tool-wrapper.sh
catalina-tasks.xml configtest.sh           migrate.bat         startup.bat          version.bat
ciphers.bat       daemon.sh               migrate.sh          startup.sh           version.sh
ciphers.sh        digest.bat              setclasspath.bat   tomcat-juli.jar

[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# ./tomcat/bin/startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Tomcat started.
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# netstat -antlpu | grep java
tcp6      0      0 127.0.0.1:8005      :::*                  LISTEN      34461/java
tcp6      0      0 :::8080            :::*                  LISTEN      34461/java
[root@tomcatsrv1 opt]#
```

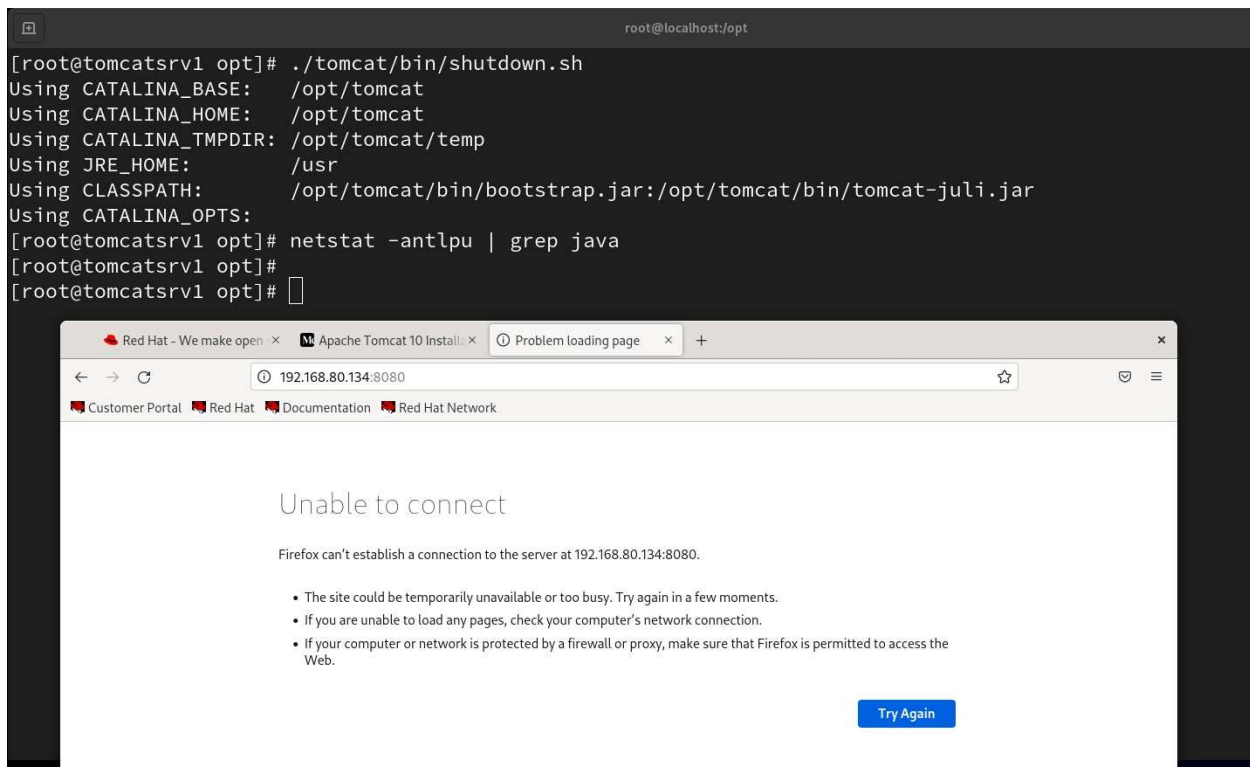

[Type here]

12)after starting service Testing Tomcat web server access by ip and port no.



The screenshot shows a web browser window with the Apache Tomcat/10.1.55 interface. The address bar shows the URL `192.168.80.134:8080`. The page has a navigation bar with links: Home, Documentation, Configuration, Examples, Wiki, Mailing Lists, and a Find Help button. The main content area features a green banner with the text "If you're seeing this, you've successfully installed Tomcat. Congratulations!". Below this is a section for "Recommended Reading" with links to "Security Considerations How-To", "Manager Application How-To", and "Clustering/Session Replication How-To". There are also buttons for "Server Status", "Manager App", and "Host Manager". A "Developer Quick Start" section includes links for "Tomcat Setup", "First Web Application", "Realms & AAA", "JDBC DataSources", "Examples", "Servlet Specifications", and "Tomcat Versions". Three yellow boxes provide additional information: "Managing Tomcat" (security, users, release notes, changelog, migration guide, security notices), "Documentation" (Tomcat 10.1 Documentation, Tomcat 10.1 Configuration, Tomcat Wiki, additional configuration information, and links to bug database, JavaDocs, and GitHub repository), and "Getting Help" (FAQ and Mailing Lists, tomcat-announce, tomcat-users, taglibs-user, tomcat-dev).

#shutdown.sh to stop the service no access form browser



The screenshot shows a terminal window and a web browser window. The terminal window, titled "root@localhost:opt", shows the execution of the `./tomcat/bin/shutdown.sh` command. The output displays the configuration for the shutdown process, including `CATALINA_BASE`, `CATALINA_HOME`, `CATALINA_TMPDIR`, `JRE_HOME`, `CLASSPATH`, and `CATALINA_OPTS`. The terminal prompt is then `netstat -antlpu | grep java`. The browser window, titled "Problem loading page", shows a "Unable to connect" error message. The message states: "Firefox can't establish a connection to the server at 192.168.80.134:8080." It lists three possible reasons: the site is temporarily unavailable, the network connection is not working, or the firewall/proxy is blocking access. A "Try Again" button is visible at the bottom right of the error message.

[Type here]

13) Creating service file

vi /etc/systemd/system/tomcat.service to start, stop

service by #systemctl

*give correct version and path of java package

```
root@localhost:opt — /usr/bin/vim /etc/systemd/system/tomcat.service
[Unit]
Description=Apache Tomcat Web Application Container
After=network.target
[Service]
Type=forking

User=tomcat
Group=tomcat

Environment="JAVA_HOME=/usr/lib/jvm/java-11-openjdk-11.0.18.0.10-2.el9_1.x86_64"
Environment="CATALINA_PID=/opt/tomcat/temp/tomcat.pid"
Environment="CATALINA_HOME=/opt/tomcat"
Environment="CATALINA_BASE=/opt/tomcat"
Environment="CATALINA_OPTS=-Xms512M -Xmx1024M -server -XX:+UseParallelGC"
Environment="JAVA_OPTS=-Djava.awt.headless=true -Djava.security.egd=file:/dev/./urandom"

ExecStart=
ExecStop=
Restart=java-11-openjdk-11.0.18.0.10-2.el9_1.x86_64
[Install]
WantedBy:jre-11
~ jre-11-openjdk-11.0.18.0.10-2.el9_1.x86_64
~ jre-openjdk
~
-- INSERT
```

*Check your actual path using #ls /usr/lib/jvm

```
root@localhost:opt
[root@tomcatsrv1 opt]# ls /usr/lib/jvm
java-11-openjdk-11.0.18.0.10-2.el9_1.x86_64  jre-11  jre-11-openjdk-11.0.18.0.10-2.el9_1.x86_64
jre  jre-11-openjdk
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# vi /etc/systemd/system/tomcat.service
[root@tomcatsrv1 opt]# cat /etc/systemd/system/tomcat.service
[Unit]
Description=Apache Tomcat Web Application Container
After=network.target
[Service]
Type=forking

User=tomcat
Group=tomcat

Environment="JAVA_HOME=/usr/lib/jvm/java-11-openjdk-11.0.18.0.10-2.el9_1.x86_64"
Environment="CATALINA_PID=/opt/tomcat/temp/tomcat.pid"
Environment="CATALINA_HOME=/opt/tomcat"
Environment="CATALINA_BASE=/opt/tomcat"
Environment="CATALINA_OPTS=-Xms512M -Xmx1024M -server -XX:+UseParallelGC"
Environment="JAVA_OPTS=-Djava.awt.headless=true -Djava.security.egd=file:/dev/./urandom"

ExecStart=/opt/tomcat/bin/startup.sh
ExecStop=/opt/tomcat/bin/shutdown.sh
```

[Type here]

14)after creating tomcat.service file :- reload daemon and stop, start the service by #systemctl cmd

```
root@localhost:/opt
[root@tomcatsrv1 opt]# systemctl daemon-reload
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# systemctl start tomcat
[root@tomcatsrv1 opt]# systemctl enable tomcat
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# systemctl status tomcat
● tomcat.service - Apache Tomcat Web Application Container
   Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-05-29 16:56:58 IST; 5min ago
     Main PID: 35520 (java)
       Tasks: 29 (limit: 10820)
      Memory: 150.8M
         CPU: 6.837s
    CGroup: /system.slice/tomcat.service
            └─35520 /usr/lib/jvm/java-11-openjdk-11.0.18.0.10-2.el9_1.x86_64/bin/java -Djava.util.logging.co

May 29 16:56:58 tomcatsrv1.radical.com systemd[1]: Starting Apache Tomcat Web Application Container...
May 29 16:56:58 tomcatsrv1.radical.com startup.sh[35513]: Tomcat started.
May 29 16:56:58 tomcatsrv1.radical.com systemd[1]: Started Apache Tomcat Web Application Container.
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# netstat -antlpu | grep java
tcp6      0      0 127.0.0.1:8005          :::*           LISTEN     35520/java
tcp6      0      0 :::8080                 :::*           LISTEN     35520/java
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]#
```

15)Configure Tomcat to Use Port 8083

Edit Tomcat's `server.xml` to change the default HTTP port:

```
Edit this file vi /opt/tomcat/conf/server.xml
```

*Before editing take backup in /tmp/

```
root@localhost:/opt
[root@tomcatsrv1 opt]# cp /opt/tomcat/conf/server.xml /tmp/server.xml.bkp
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# vi /opt/tomcat/conf/server.xml
```

[Type here]

```
and responses are returned. Documentation at :
HTTP Connector: /docs/config/http.html
AJP Connector: /docs/config/ajp.html
Define a non-SSL/TLS HTTP/1.1 Connector on port 8080
-->
<Connector port="8083" protocol="HTTP/1.1"
    connectionTimeout="20000"
    redirectPort="8443"
    maxParameterCount="1000"
/>
<!-- A "Connector" using the shared thread pool-->
<!--
<Connector executor="tomcatThreadPool"
    port="8080" protocol="HTTP/1.1"
    connectionTimeout="20000"
    redirectPort="8443"/>
```

*service stop and start is required

16)changed the Tomcat default port from 8080 to 8083 and accessed it through browser to verify is it running on new configured port

The screenshot shows a terminal window and a web browser. The terminal window displays the following commands and output:

```
[root@tomcatsrv1 opt]# vi /opt/tomcat/conf/server.xml
[root@tomcatsrv1 opt]# netstat -antlpu | grep java
tcp6      0      0 127.0.0.1:8005        :::*           LISTEN     35937/java
tcp6      0      0 :::8083              :::*           LISTEN     35937/java
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]#
```

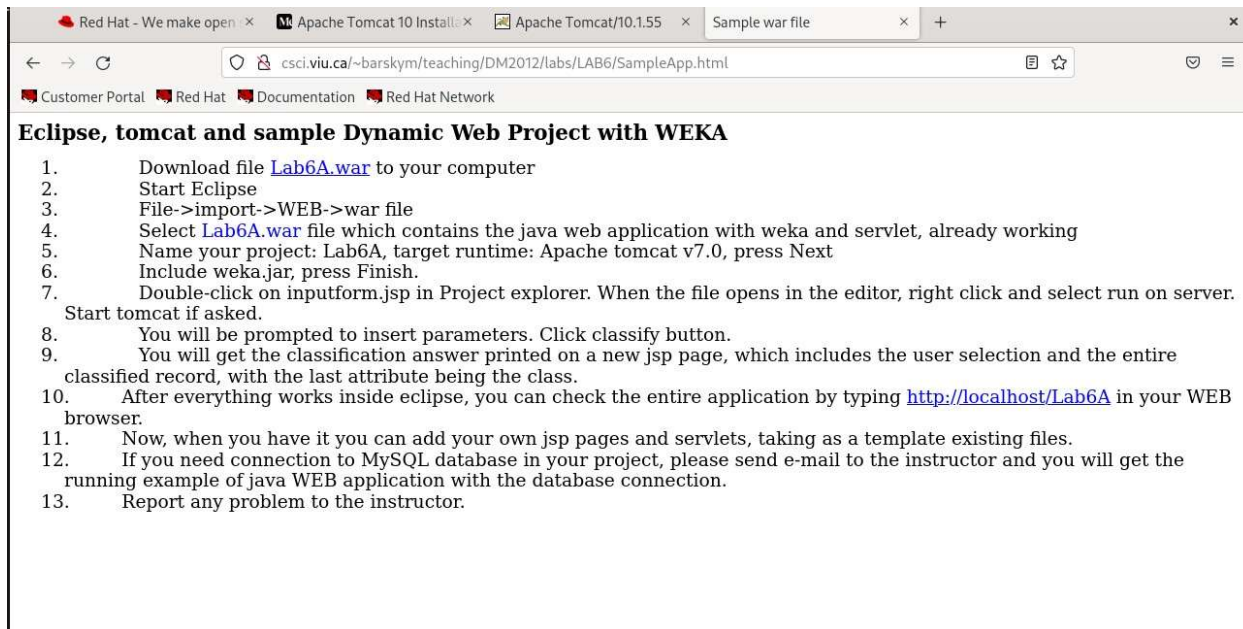
The web browser shows the Apache Tomcat/10.1.55 installation page. The page includes a green banner with the text "If you're seeing this, you've successfully installed Tomcat. Congratulations!". Below the banner, there is a section for "Recommended Reading" with links to "Security Considerations How-To", "Manager Application How-To", and "Clustering/Session Replication How-To". There are also buttons for "Server Status", "Manager App", and "Host Manager". The page also features a "Developer Quick Start" section with links to "Tomcat Setup", "First Web Application", "Realms & AAA", "JDBC DataSources", "Examples", "Servlet Specifications", and "Tomcat Versions".

[Type here]

17) Deploy Your Application as Lab6A.war

Copy your WAR file into Tomcat's webapps directory:

<http://csci.viu.ca/~barskym/teaching/DM2012/labs/LAB6/SampleApp.html>

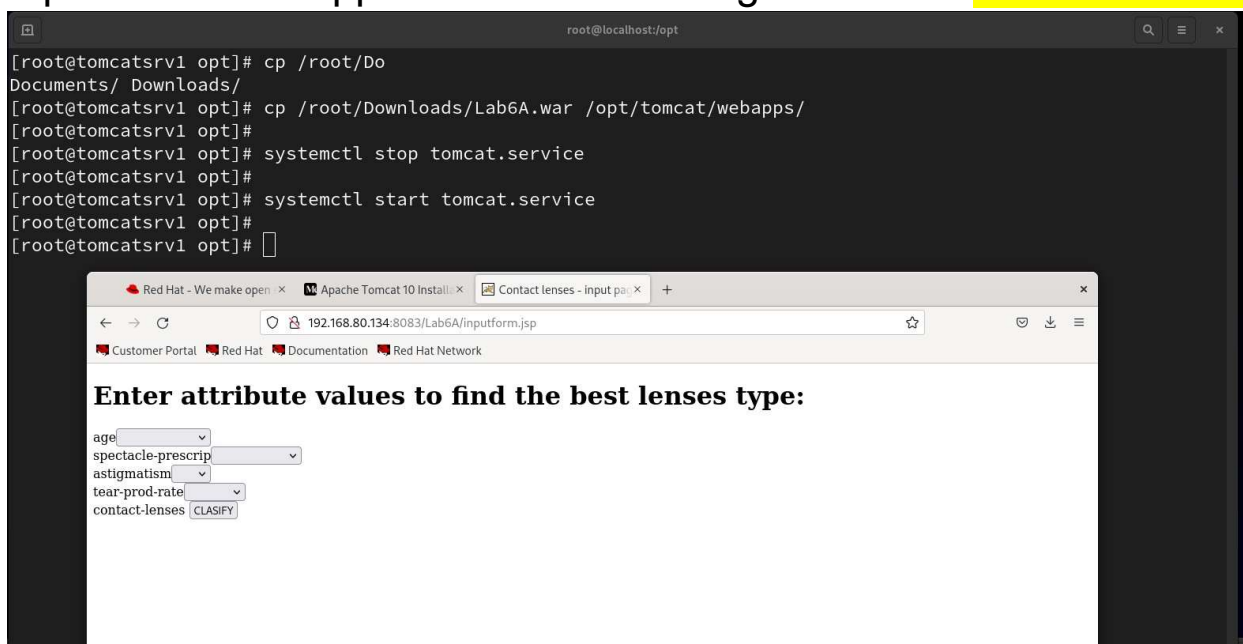


The screenshot shows a web browser window with the address bar displaying `csci.viu.ca/~barskym/teaching/DM2012/labs/LAB6/SampleApp.html`. The page title is "Eclipse, tomcat and sample Dynamic Web Project with WEKA". The page contains a numbered list of 13 steps for deploying the application:

1. Download file [Lab6A.war](#) to your computer
2. Start Eclipse
3. File->import->WEB->war file
4. Select [Lab6A.war](#) file which contains the java web application with weka and servlet, already working
5. Name your project: Lab6A, target runtime: Apache tomcat v7.0, press Next
6. Include weka.jar, press Finish.
7. Double-click on inputform.jsp in Project explorer. When the file opens in the editor, right click and select run on server. Start tomcat if asked.
8. You will be prompted to insert parameters. Click classify button.
9. You will get the classification answer printed on a new jsp page, which includes the user selection and the entire classified record, with the last attribute being the class.
10. After everything works inside eclipse, you can check the entire application by typing <http://localhost/Lab6A> in your WEB browser.
11. Now, when you have it you can add your own jsp pages and servlets, taking as a template existing files.
12. If you need connection to MySQL database in your project, please send e-mail to the instructor and you will get the running example of java WEB application with the database connection.
13. Report any problem to the instructor.

after download Lab6A.war is in /root/Downloads copy to

/opt/tomcat/webapps and access through browser **192.168.80.134:8083/Lab6A**



The screenshot shows a terminal window and a web browser window. The terminal window displays the following commands and output:

```
[root@tomcatsrv1 opt]# cp /root/Do
Documents/ Downloads/
[root@tomcatsrv1 opt]# cp /root/Downloads/Lab6A.war /opt/tomcat/webapps/
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# systemctl stop tomcat.service
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# systemctl start tomcat.service
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]#
```

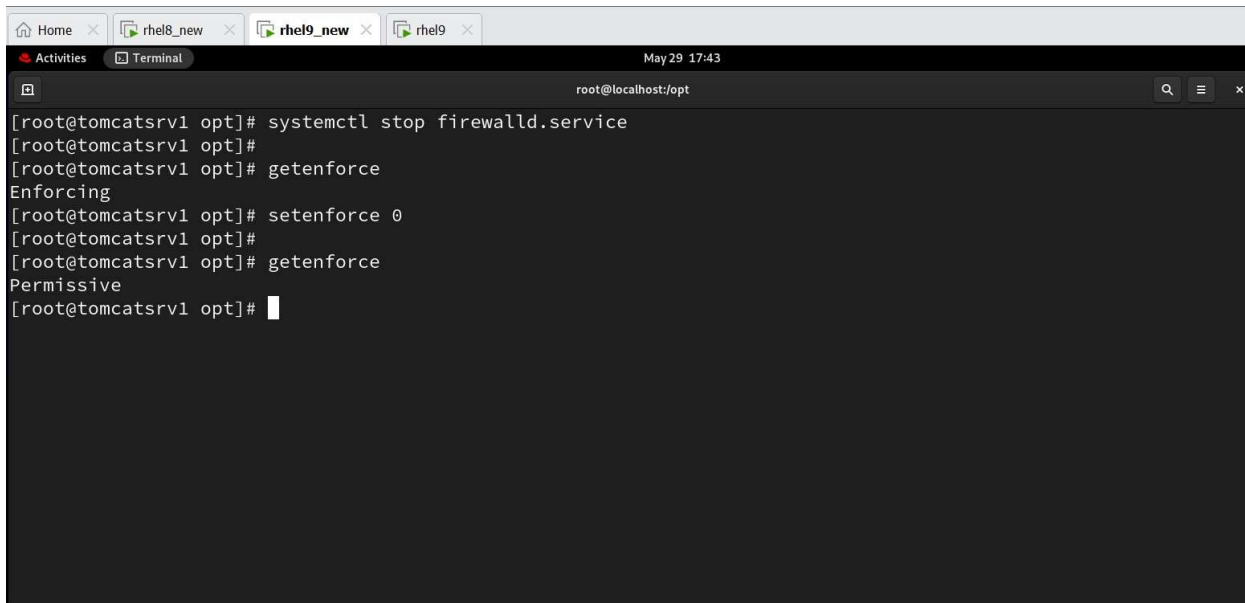
The web browser window shows the address bar with `192.168.80.134:8083/Lab6A/inputform.jsp`. The page title is "Enter attribute values to find the best lenses type:". The page contains a form with the following attributes:

- age
- spectacle-prescrip
- astigmatism
- tear-prod-rate
- contact-lenses

There is a "CLASSIFY" button at the bottom of the form.

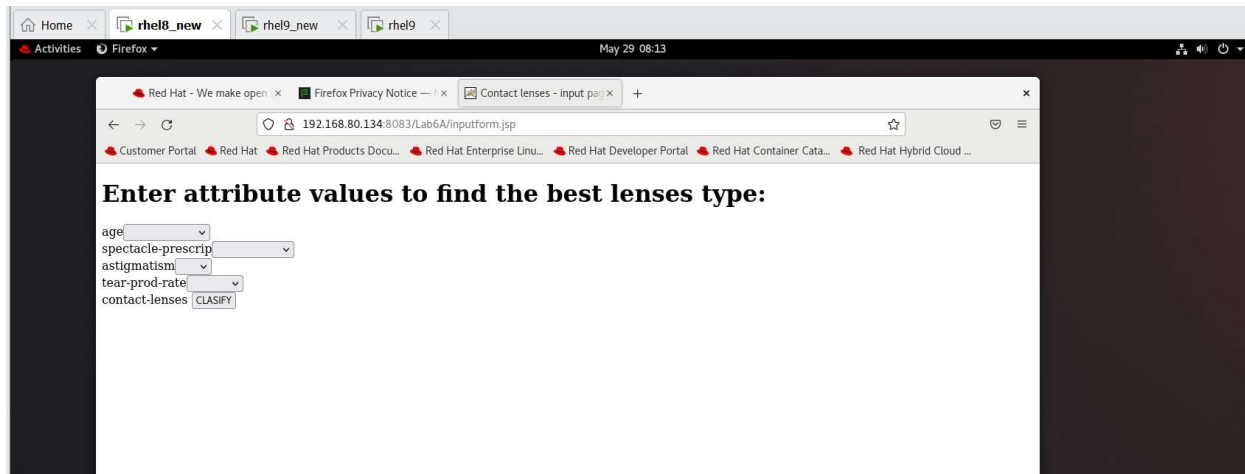
[Type here]

18)to test from client side stop firewalld and selinux service



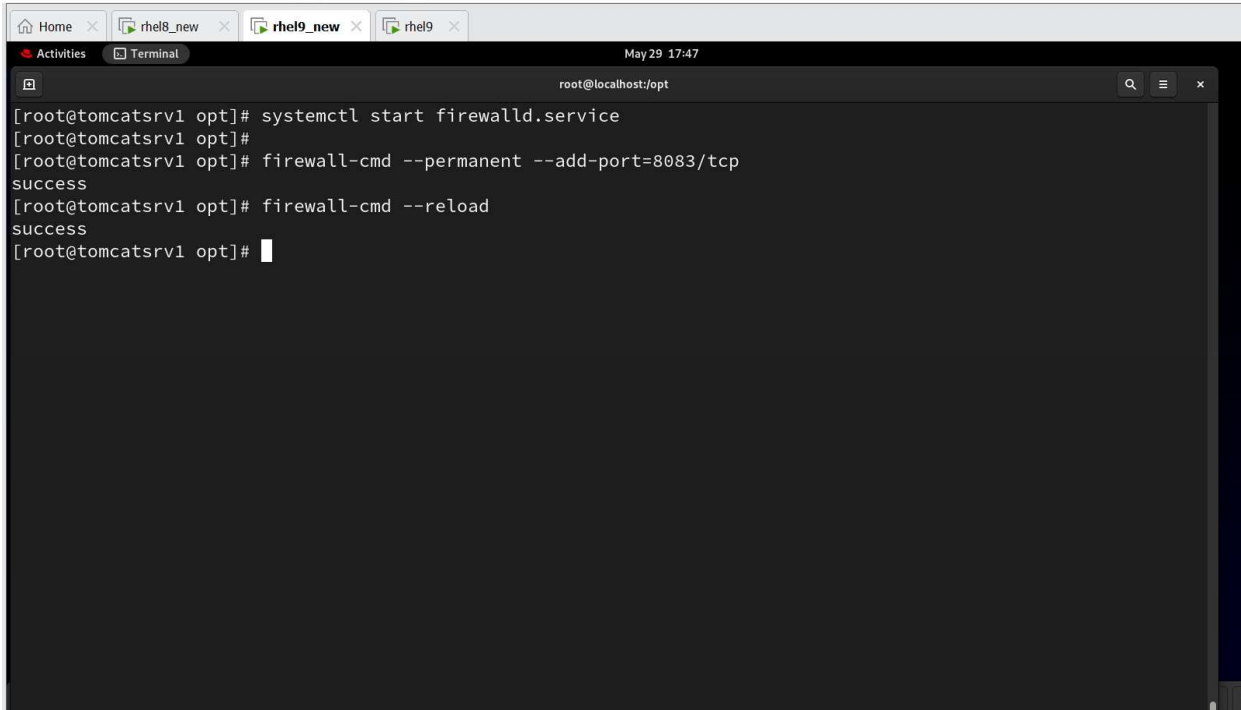
```
[root@tomcatsrv1 opt]# systemctl stop firewalld.service
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# getenforce
Enforcing
[root@tomcatsrv1 opt]# setenforce 0
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# getenforce
Permissive
[root@tomcatsrv1 opt]#
```

19) Accessible from client side



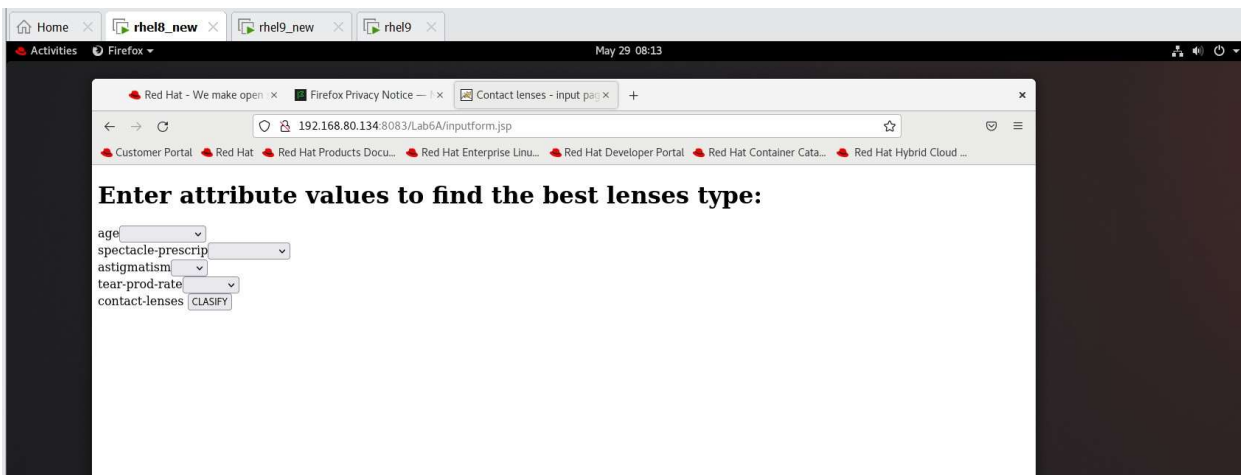
[Type here]

20) Configured firewall rules to permit Tomcat traffic on port 8083



A terminal window titled 'root@localhost:opt' showing the following commands and output:

```
[root@tomcatsrv1 opt]# systemctl start firewalld.service
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# firewall-cmd --permanent --add-port=8083/tcp
success
[root@tomcatsrv1 opt]# firewall-cmd --reload
success
[root@tomcatsrv1 opt]#
```



Conclusion

Apache HTTP Server and Apache Tomcat were successfully installed and configured on RHEL 9. The application deployment was verified through browser access and service validation commands. This setup demonstrates a standard enterprise-level Java web application deployment environment.